Probing Spin Relaxometry in Beesworks® Beeswax with Pulsed NMR

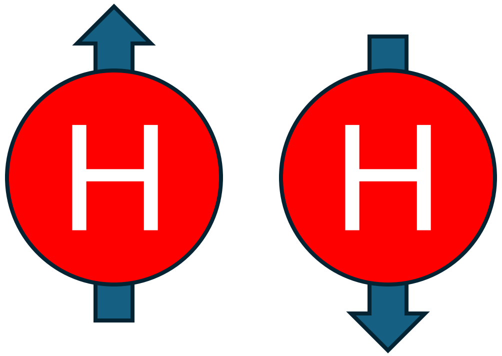
Ethan Suresh and Max Misterka

8.13 Experimental Physics I, Spring 2026

Nuclear Magnetism

Electrons, protons, neutrons contain an intrinsic property known as **spin;** superposition of spin up and down

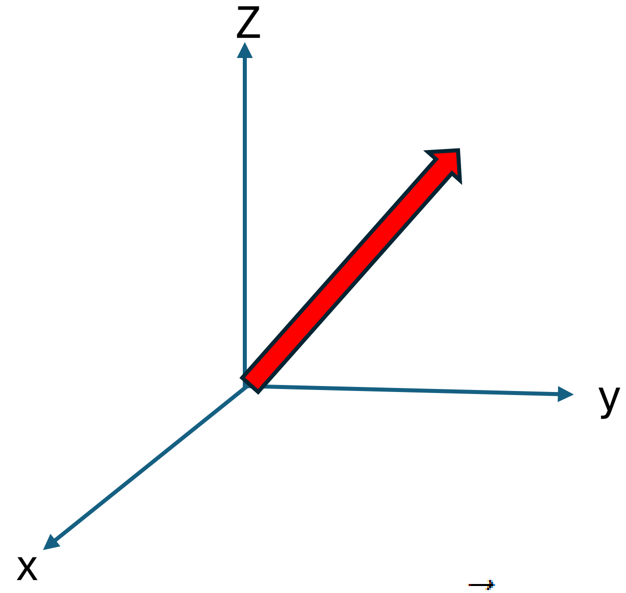
Quantum Representation



S = 1/2 S = -1/2

Magnetic Moment (J/T): $\mu = \frac{\gamma \hbar}{2}$

Classical Representation

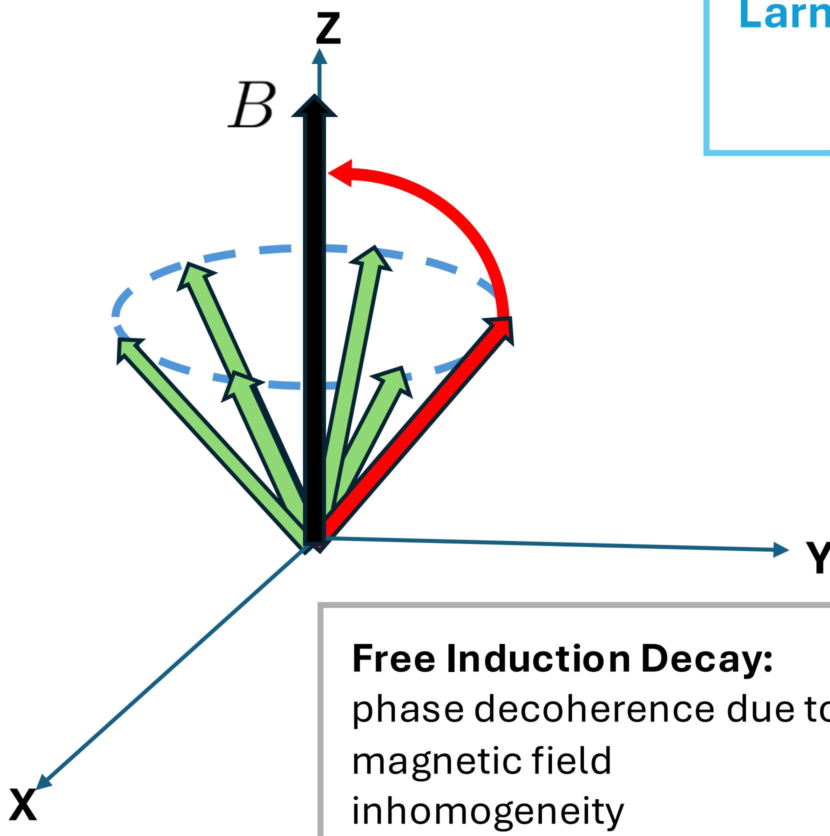


Net Magnetization $\vec{\mathrm{M}} = N \langle \vec{\mu} \rangle$

Sources:

NMR: What happens to the spin population in an external magnetic field?

$$H_{spin} = -\vec{\mu} \cdot \vec{\mathrm{B}}$$



Larmor Precession:

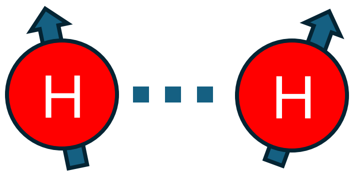
$$\omega_L = \gamma B$$

Free Induction Decay: phase decoherence due to magnetic field inhomogeneity

Spin-Lattice Relaxation (T1): Spin population converges to direction of B

$$M_z(t) = M_0(1 - \exp(-t/T_1))$$

Spin-Spin Relaxation (T2): neighboring spin-spin dipole interactions exchange energy, causing different local precession rates



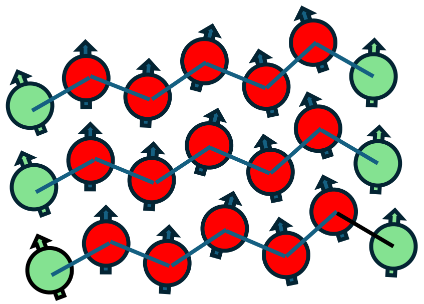
$$M_{xy}(t) = M_0 \exp(-t/T_2))$$

Sources:

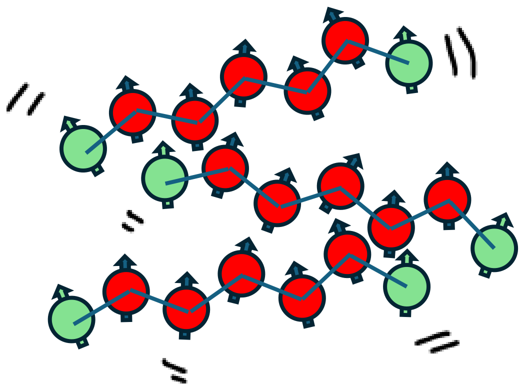
Relaxometry of Beeswax: T2 increases with temperature due to decreasing chemical anisotropy

- **Beesworks® Beeswax:** composite organic material made of esters, hydrocarbons, fatty acids (long chains of C, O, H), each with their own relaxation times

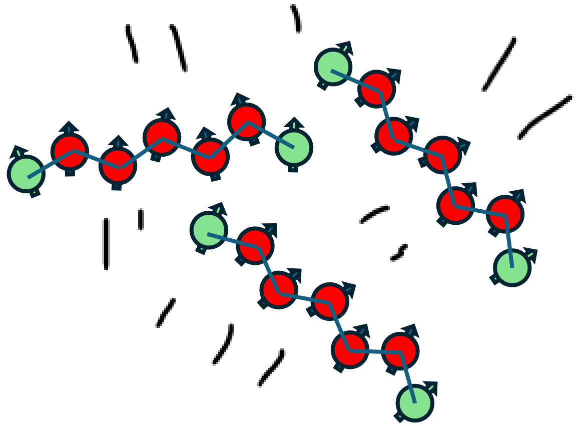
Solid (Crystalline) Beeswax
(< 40 C)



Semicrystalline Beeswax
(40-60 C)



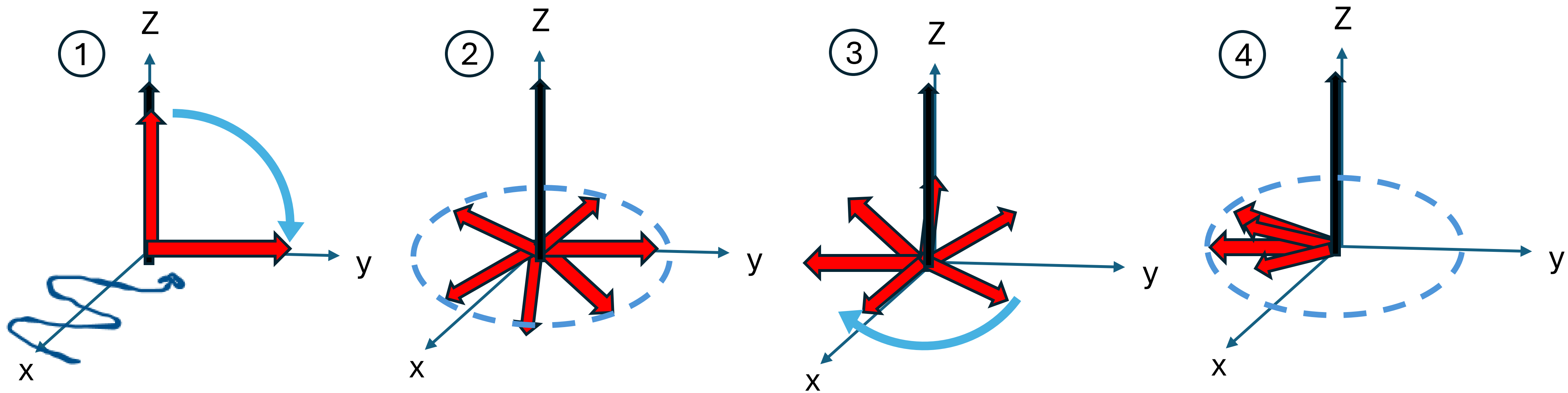
Liquid Beeswax (60-70 C)

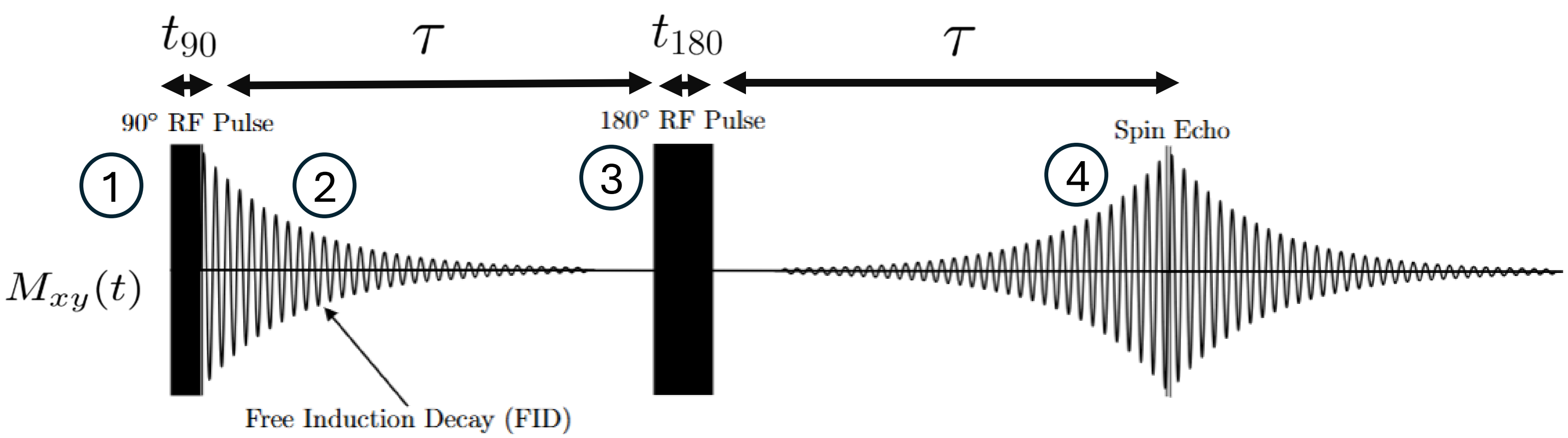


Hypothesis: as temperature increases, dipole interaction strength decreases relative to thermal energy and thus chemical environments become more uniform; less phase decoherence

Sources:

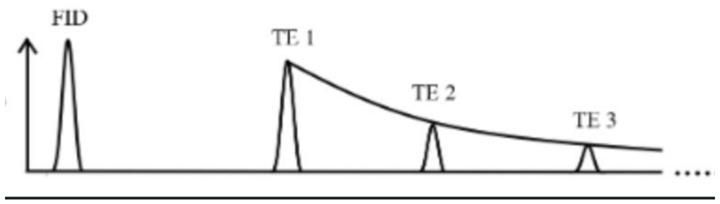
NMR: Using a Spin Echo For Measuring T2, not T2*





Measuring spin echo amplitude at different times yields T2!

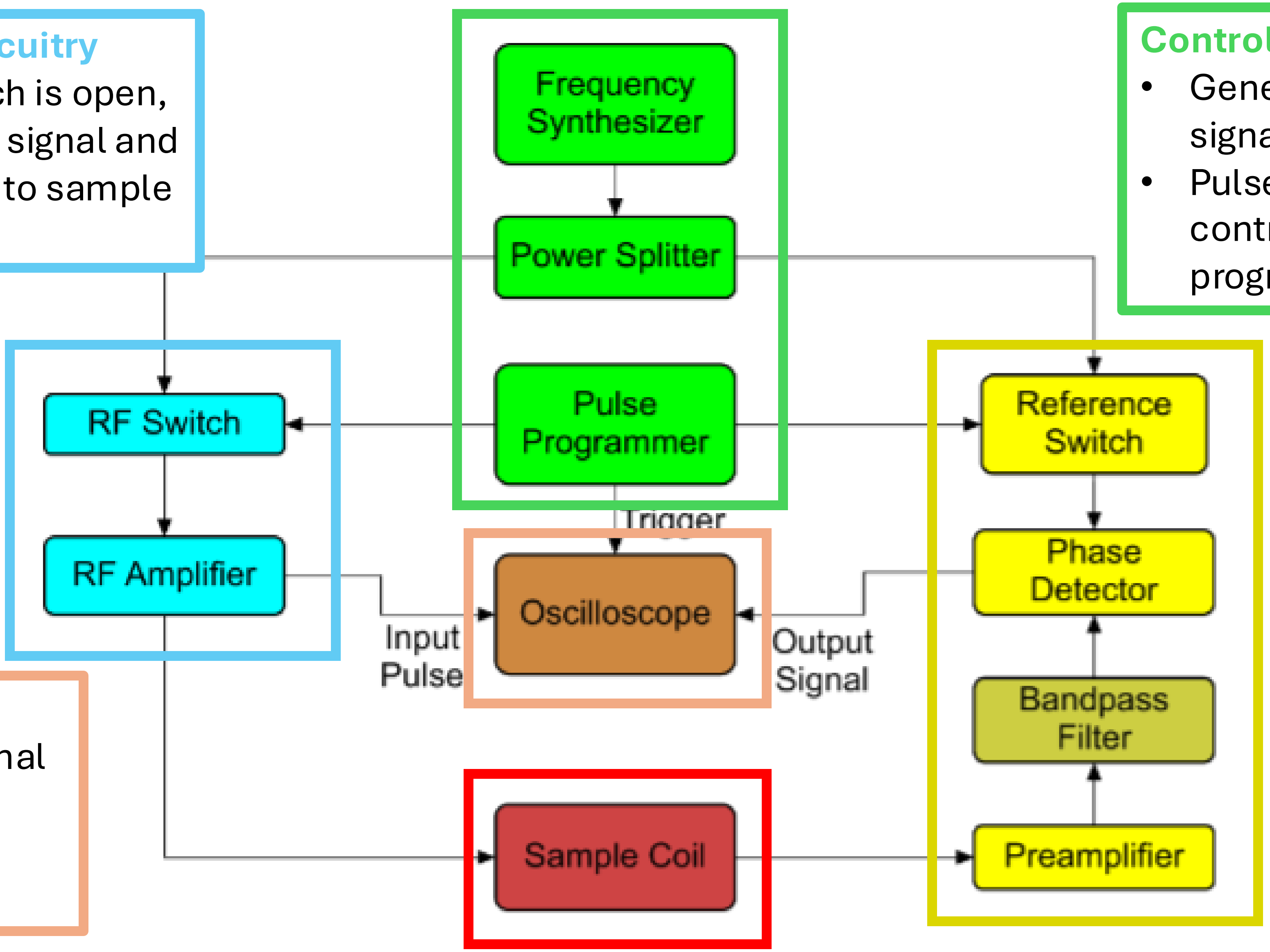
$$M_{xy}(t) = M_0 \exp(-2\tau/T_2))$$



Sources: Jlab Guide, [T2 Mapping - Quantitative MRI mOOC](#) (Image)

Apparatus for pulsed NMR sequences

Transmission Circuitry
- When the switch is open, amplifies input signal and transmits B1(t) to sample coil



Control Circuitry:
- Generate oscillating RF signal near resonance
- Pulse duration (t90, t180) controlled by pulse programmer

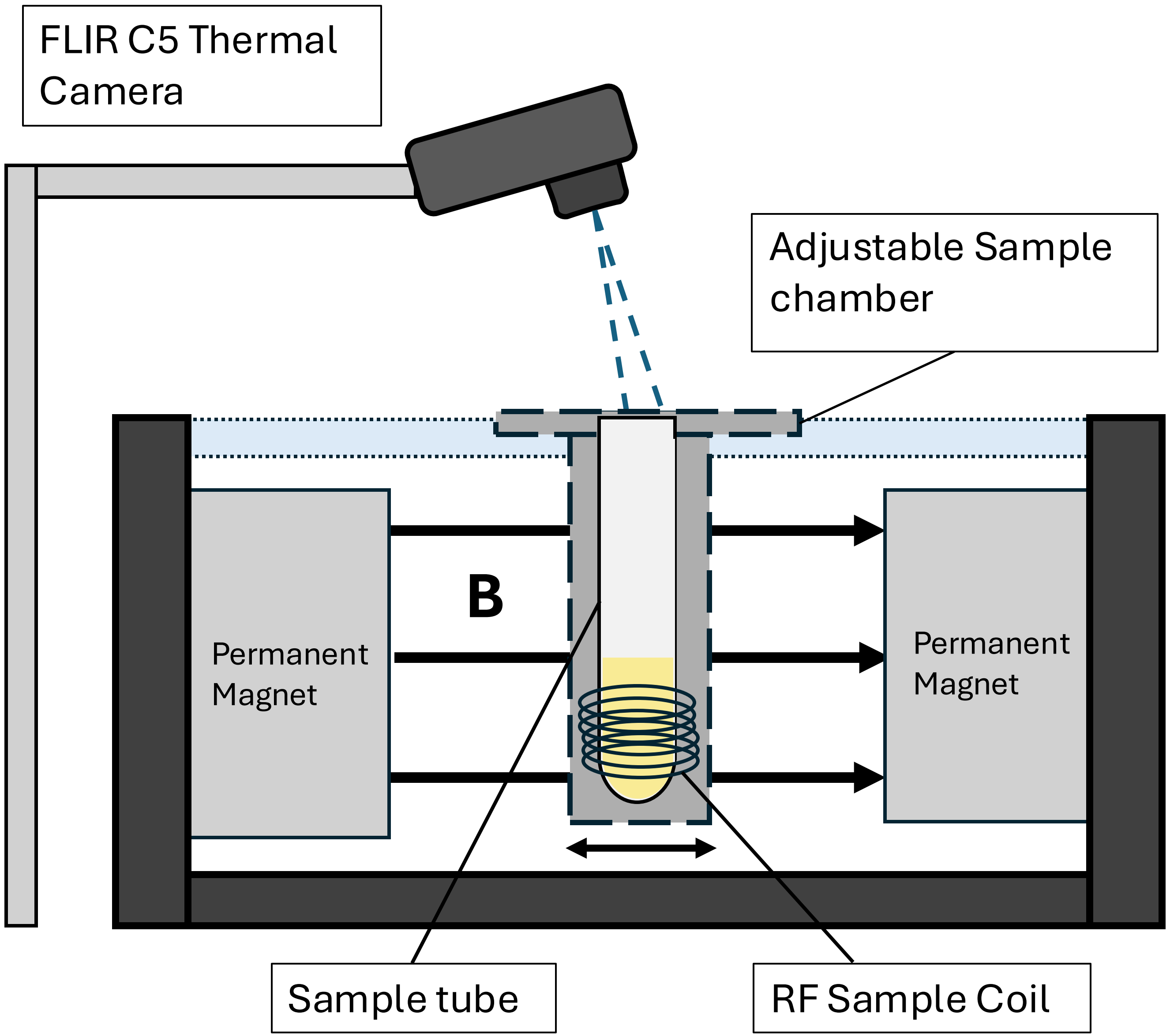
Signal Detection
- Amplify response signal detected by coil
- Superimpose response signal with reference signal

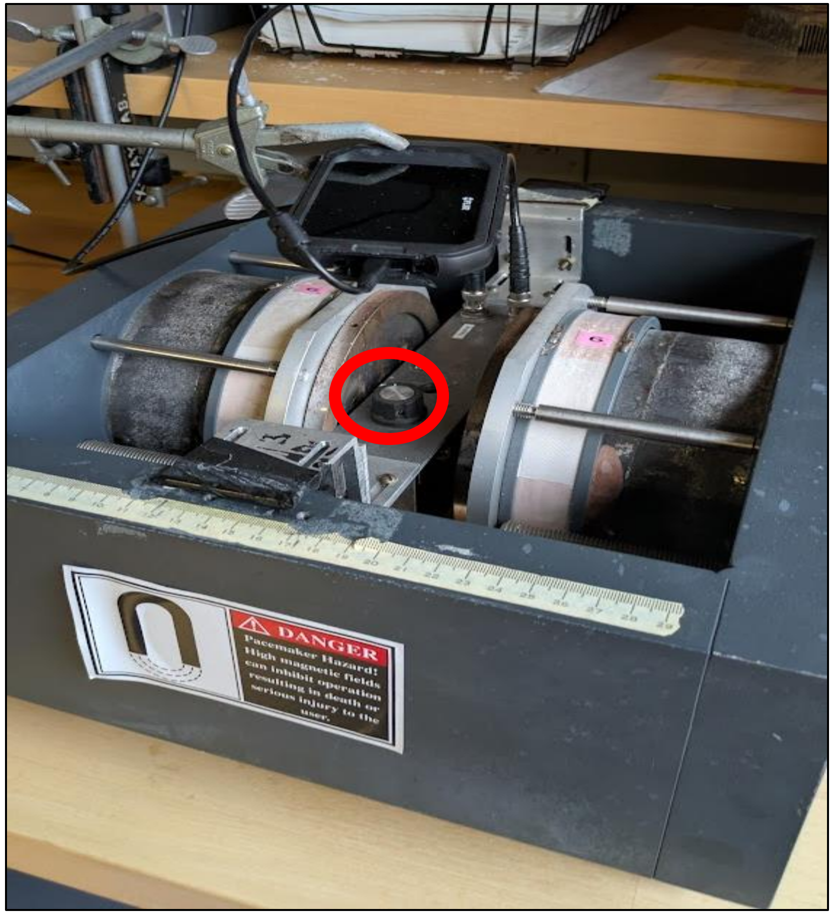
Oscilloscope
- Display beat signal with frequency

$$\omega_L - \omega_I$$

Sources:

NMR Sample Apparatus





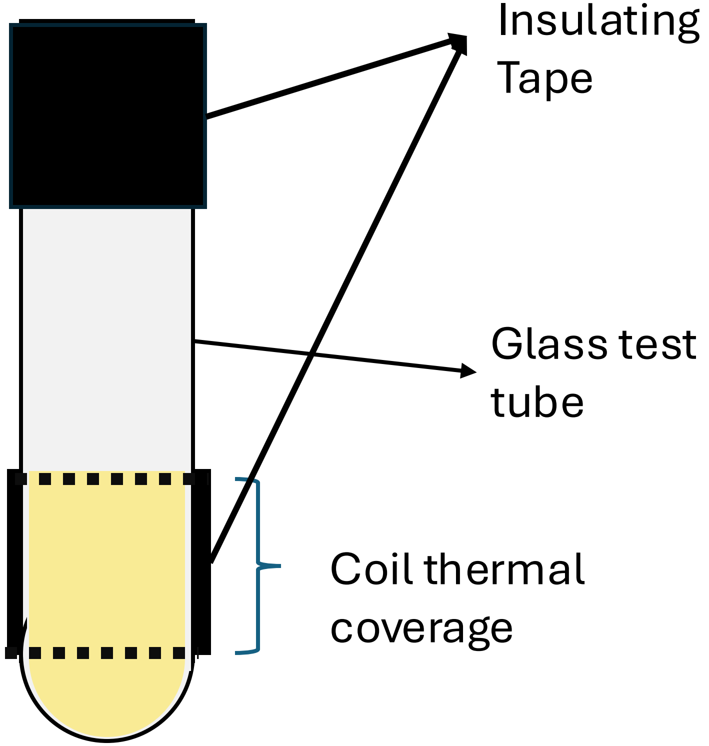
Tuning Capacitor

Sources:

Anatomy of NMR sample and Temperature visualization

Beesworks ® Beeswax sample

Sample and Tube materials

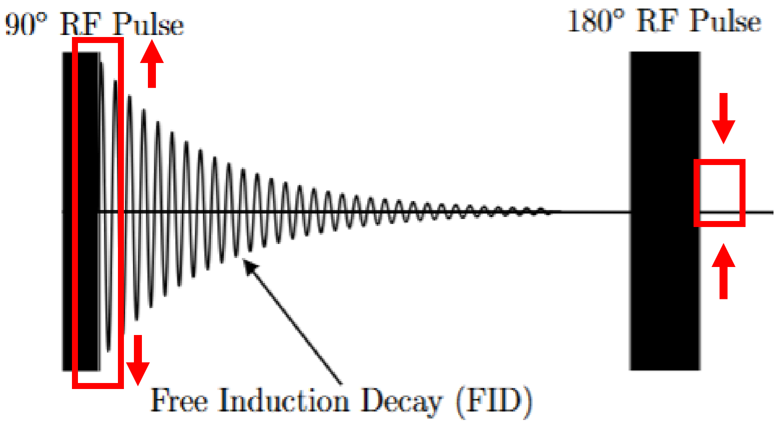


Thermal Camera visualization

Experimental Procedure for measuring T2 in Beeswax

0. Initial calibration:

PW1, PW2, tuning capacitor calibrated for a 90-180 spin echo

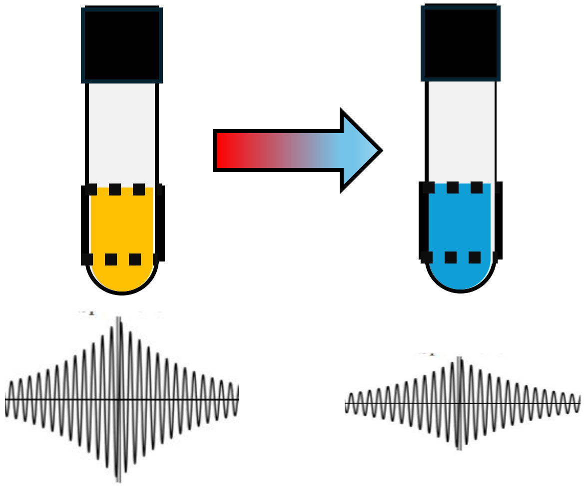


1. Beeswax Heating

Sample heated to past 70 C using a hot plate

2. Spin Echo measurements

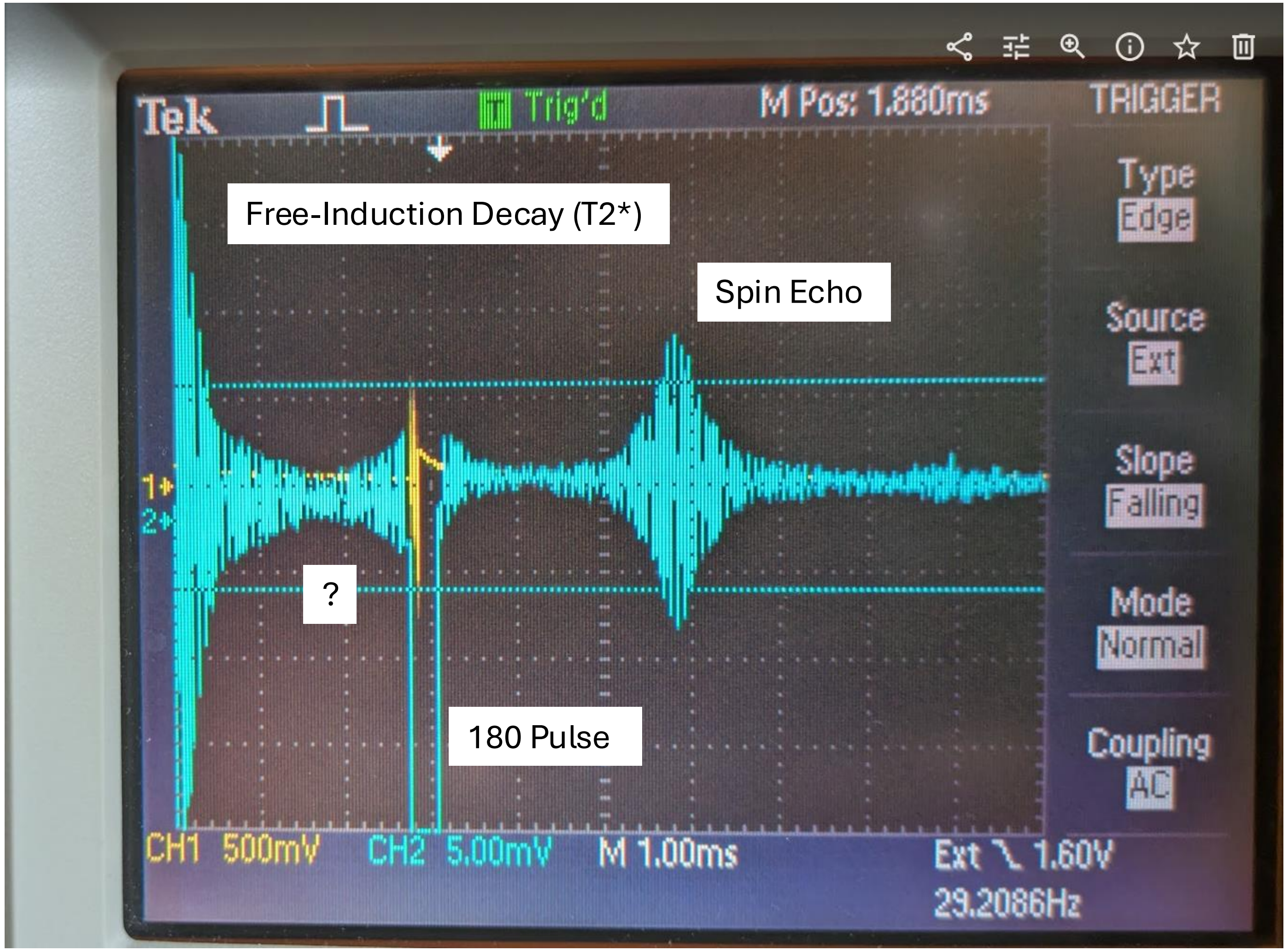
- Sample temperature T_cam measured using thermal camera
- As sample cools from 70 to 45 C, spin echo amplitude is recorded using oscilloscope and phone camera



Repeat steps 1 and 2 for five different delay times **τ**: 2, 6, 10 16, 18 ms

Sources:

Example Spin-Echo Signal for measuring T2



Apparatus Settings

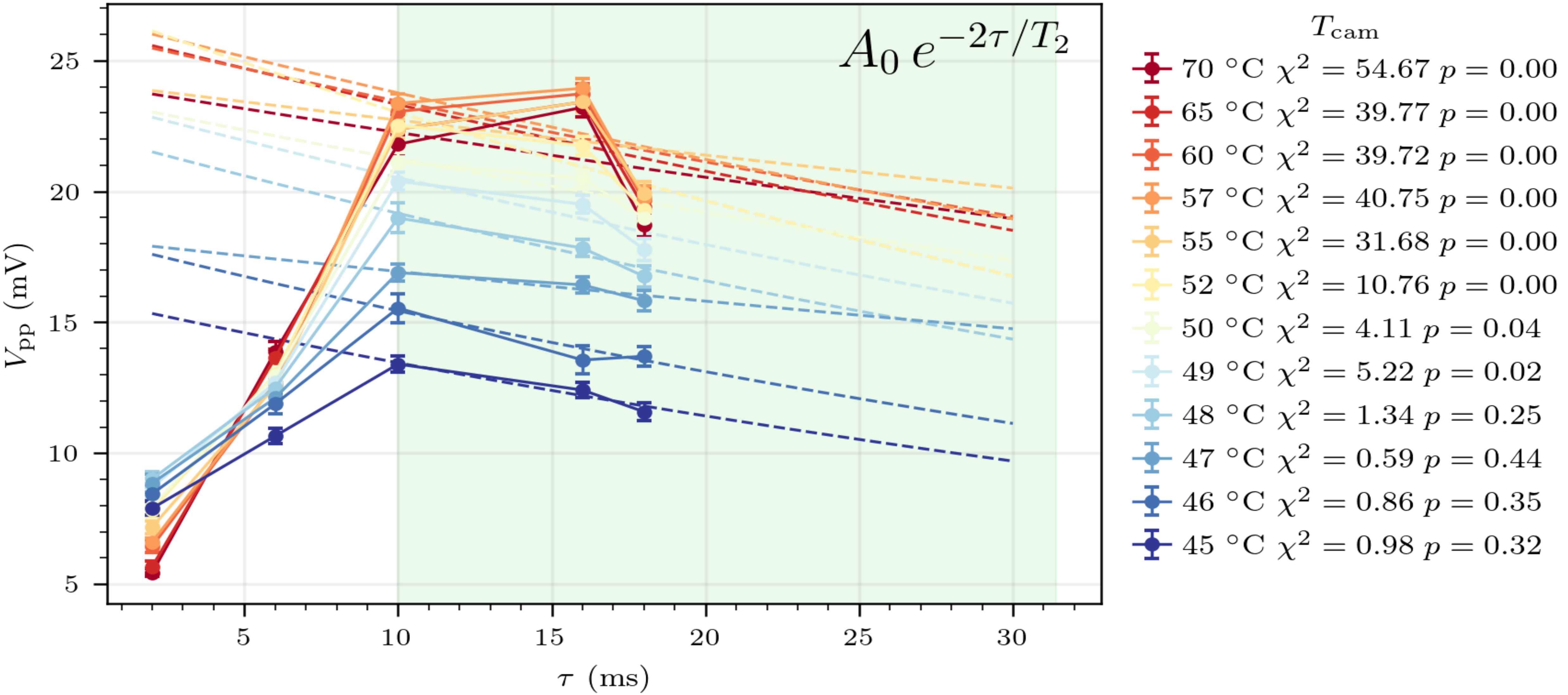
- **Tau:**
- **PW1 (t90)**: 31 us +/- 2
- **PW2: (180):** 62 us +/- 5
- **Averaging:** 16 (necessary to filter out noise)
- Repeat time was not held constant but was long compared to tau (> 180 ms)

Spin Echo Amplitude Measurement

- Peak-to-peak spin echo amplitude obtained digitally by measuring in the recording

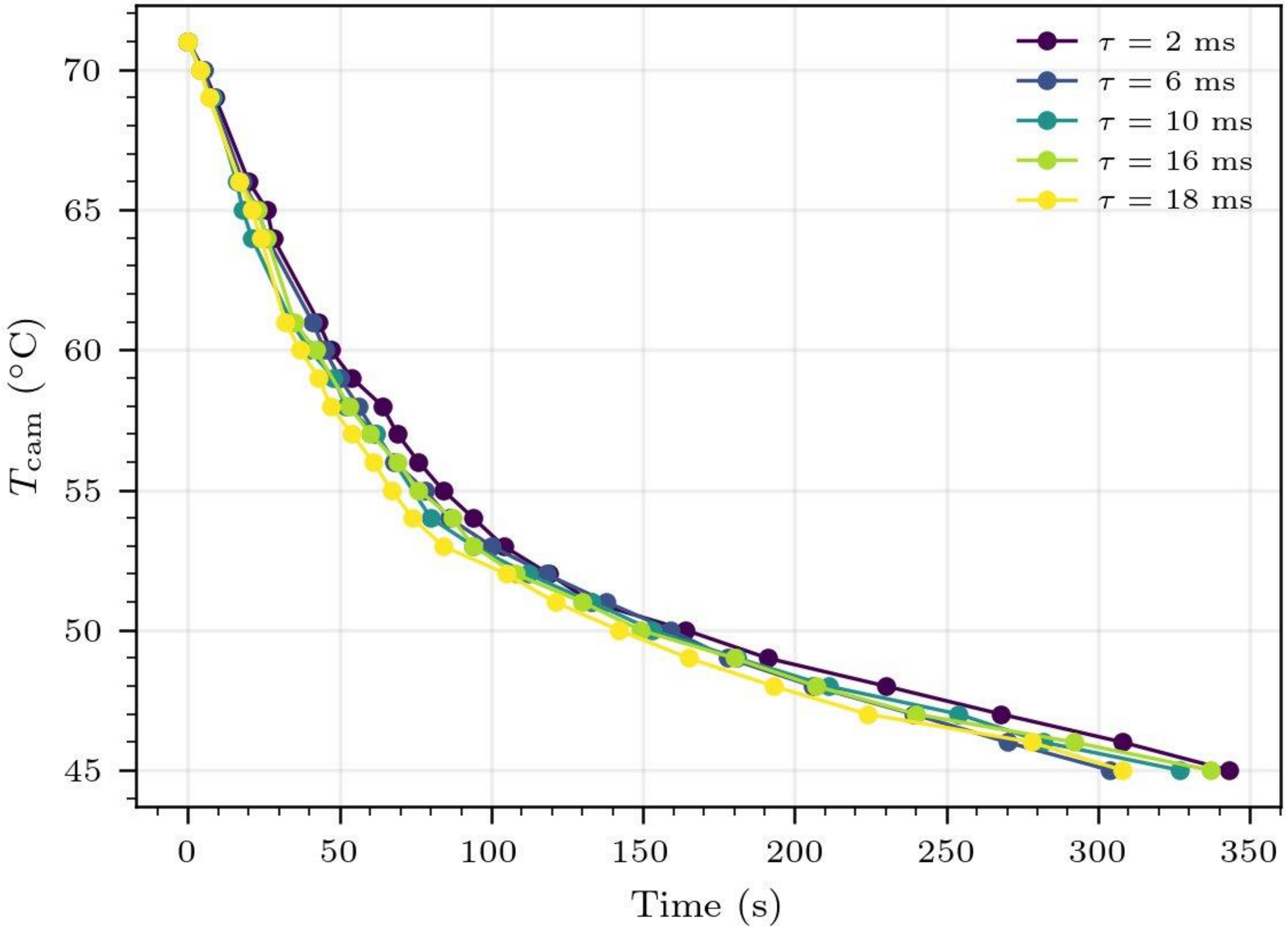
Sources:

Spin-Echo Amplitudes do not fit decaying exponential trend



Sources:

Quantifying uncertainty in T_{cam} with cooling rate



Numerical Derivative Estimate

$$\frac{dT}{dt}(T_i) = \frac{T_{i+1} - T_{i-1}}{t_{i+1} - t_{i-1}}$$

Characteristic Delay Time Estimate

$$\Delta t \approx 1\mathrm{s}$$

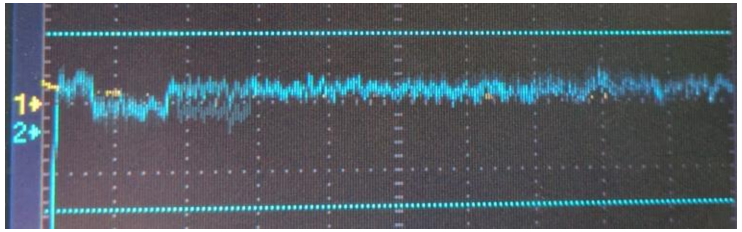
Mean Temperature Uncertainty

$$\sigma_T(T) = \Delta t \cdot \frac{1}{N_\tau} \sum_{k=1}^{N_\tau} \left| \frac{dT}{dt} \right|_k (T)$$

Sources:

Other systematic errors we didn't account for

1. Time scale (τ) limitations:
- Spin echo disappears after increasing tau past 18 ms; likely due to trigger timing and averaging
- Large systematic fit uncertainty



2. Anomalous saturation effect
- Increases fit uncertainty; fit a possibly due to influence of T_1 (spin-lattice relaxation) and repeat time; $T_1 \sim T_2$

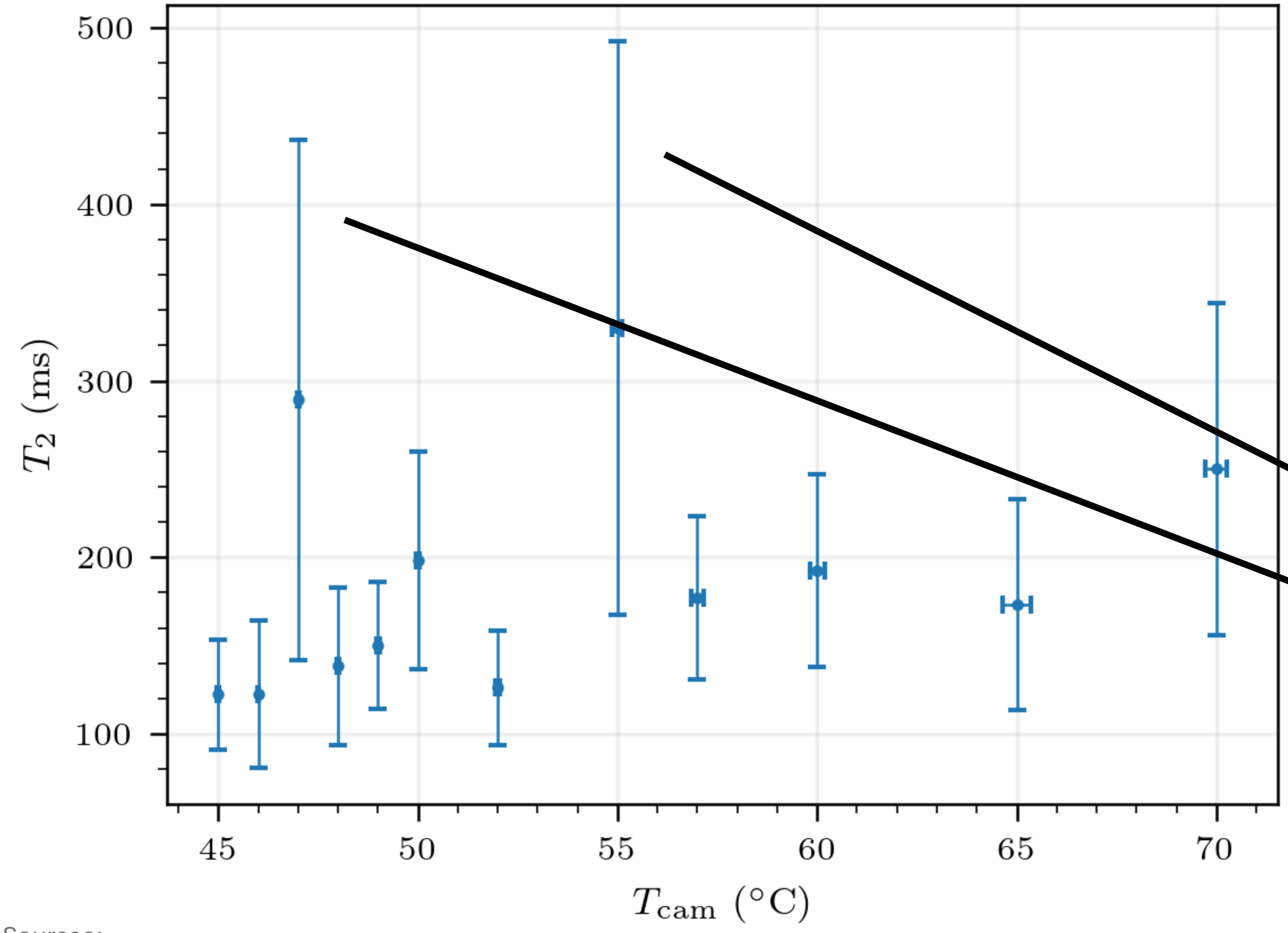
3. Non-equilibrium Cooling
- Cooling is not **quasistatic ($t_{relax} \ll t_{cool}$)**
- Multiple phases present at the same time
- Lowers T_{cam} and <u>might lower T_2</u>
- Look into insulated sample holder

4. Apparatus Errors
- Pulse width miscalibration: likely does not change relative trend
- Sample location in holder

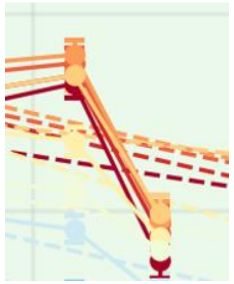
5. Temperature Calibration
- $T_{cam} < T_{real}$; systematic temperature shift
- Measure T_{real} simultaneously with T_{cam}

Sources:

Uncertain temperature dependence of T2



Higher temperatures likely have a much lower T2; systematic effect from fit

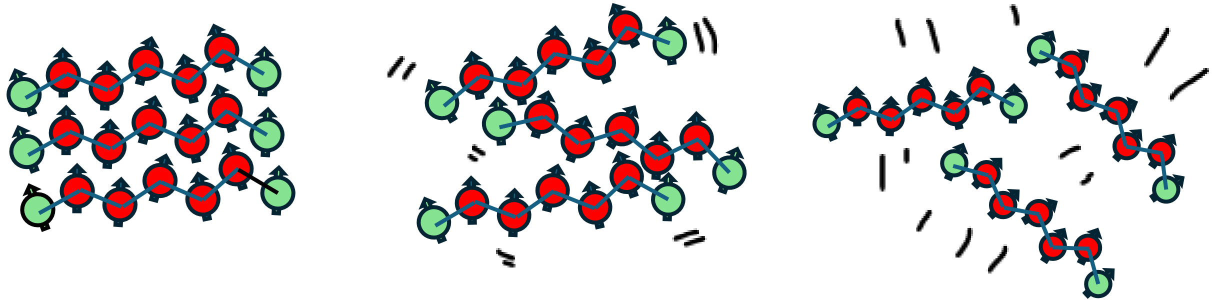


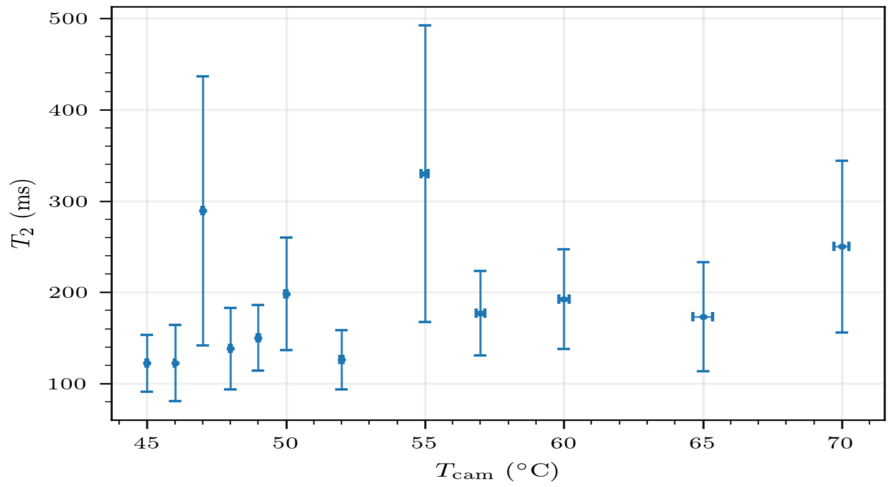
Are outliers a statistical fluke or physically motivated, eg. A phase transition?

Sources:

Conclusion: Temperature does seem have an effect on T2 relaxometry in Beesworks® beeswax, but more investigation is required to rigorously determine the nature of this effect.

- **Hypothesis: T2 increases with temperature due to decreasing chemical anisotropy**





- **Results**: Large variation in T2, but inconclusive interperation
- **Statistical Errors**: Measurement noise, different trials for different tau, fit uncertainty
- **Systematic Error:** Temperature cooling (accounted), low range of tau (unaccounted), **etc.**

Sources:

Thank you!

- **Lab partner:** Max Misterka
- **Jlab Instructional Staff:** Prof. Janet Conrad, Jixiang Yang, Simon Rothman, Vinh Tran, Atissa Banuazizi
- **JLab Technical Staff:** Dr. Sean Robinson, Dr. Aaron Pilarcik, Cory Romanov
- **Claude Code** was used to generate code for data analysis
- **ChatGPT** was used for beeswax NMR literature review

Sources: